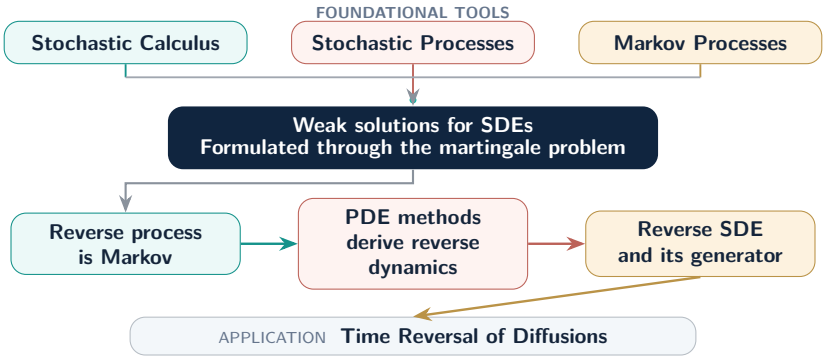
Operator semigroups

# Time Reversal of Diffusions

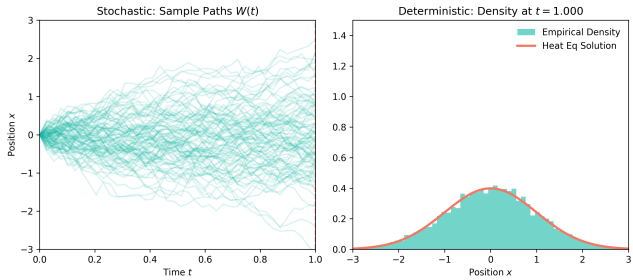
Subtitile

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# From foundations to time reversal



# Sample paths of Brownian motion




## Six independent paths

Each curve is a realization of standard Brownian motion. All paths begin at zero and evolve independently.

$$B_0^{(k)} = 0, \quad k = 1, \ldots, 6.$$

The animation loops as each trajectory unfolds.

# The Idea of Diffusion

## Start with a weighted semigroup average

For a kernel $\phi$, define

$$P_\phi f(x) := \int_0^\infty \phi(t)\, P_t f(x)\, dt.$$

Taking the exponential kernel

$$\phi_p(t) := e^{-pt}, \qquad p > 0,$$

recovers the resolvent, which is the Laplace transform of the semigroup:

$$R_p \equiv P_{\phi_p}.$$

# The exponential kernel gives the resolvent

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# Build a class from exponential kernels

Consider finite linear combinations of exponentials:

$$\mathcal{C} := \left\{ \phi : [0,\infty) \to \mathbb{R} \;\middle|\; \phi(t) = \sum_{i=1}^{n} c_i e^{-p_i t},\ c_i \in \mathbb{R},\ p_i > 0,\ n \in \mathbb{N} \right\}.$$

By linearity, every $\phi \in \mathcal{C}$ satisfies

$$\langle P_\phi f, g\rangle_\nu = \sum_{i=1}^{n} c_i \langle R_{p_i} f, g\rangle_\nu = \langle f, \widehat{P}_\phi g\rangle_\nu.$$

Linearity carries the resolvent adjoint relation to finite exponential sums.

**Previous result** For each $p > 0$, $R_p = P_{\phi_p}$.

# Density extends the identity to $C_0$

The exponential class is dense in $C_0([0,\infty))$. Thus, for each $\phi \in C_0([0,\infty))$, there is a sequence $\phi_n \in \mathcal{C}$ such that

$$\lim_{n\to\infty} \sup_{t\geq 0} \left|\phi_n(t) - \phi(t)\right| = 0.$$

Uniform approximation $\Longrightarrow$ extension beyond exponentials